

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 5. Conditional Statements

Lecture: 45. Challenge - Leap Year or Not

Section 1: Lecture Summary

This lecture comprehensively discusses the concept of a **leap year**, starting from its definition to the mathematical and historical reasoning behind it, and culminating in the implementation of a Python program to check leap years. Initially, the lecture explains why February sometimes has 29 days, defining a leap year as a year where February has 29 days, making the year 366 days long instead of 365. It then delves into how the Earth's revolution around the Sun, which takes approximately 365.242 days, necessitates the adjustment of calendar years by adding leap years.

The lecture covers two calendar systems:

- **Julian calendar:** A simple rule that every year divisible by 4 is a leap year, which causes small errors accumulating over centuries.
- **Gregorian calendar:** Introduces corrections by considering century years and leap rules involving division by 100 and 400 to more accurately reflect the Earth's revolution time.

Finally, the lecture presents the correct logical conditions to check for a leap year in Python using nested `if-elif-else` statements and demonstrates running the program interactively in a Jupyter Notebook environment. The stepwise explanation teaches students both the problem-solving approach and the practical way to implement and test the program.

Section 2: Key Concepts and Explanations

- What is a Leap Year?

- A leap year has 366 days instead of the usual 365 days.
- It occurs when February has 29 days instead of 28.
- Leap years are inserted to compensate for the fact that the Earth's orbit around the Sun takes approximately **365.242** days.

- Earth's Revolution and the Need for Leap Years

- The actual time for Earth's revolution is about 365.242 days (not exactly 365).
- Adding a quarter of a day every year (0.25) accumulates to an extra day every 4 years.
- This leads to the basic leap year rule: **every 4th year is a leap year.**

- Julian Calendar Rule (Old System)

- Every year divisible by 4 is a leap year.
- This causes an inaccuracy because the true year length is closer to 365.242, not 365.25.
- The small error (.008 days per year) accumulates resulting in an extra day approximately every 125 years.

- Gregorian Calendar Rule (Modern System)

- Leap year if the year is divisible by 4 **and** the following adjustments:
 - If the year is divisible by 100 (a century year), then it is **not** a leap year unless...
 - The year is also divisible by 400, then it **is** a leap year.
- This adjustment fixes the extra day problem accumulated by the Julian calendar.
- Formula shorthand:

$$\text{Leap Year} = (\text{year} \% 4 == 0) \wedge ((\text{year} \% 100 \neq 0) \vee (\text{year} \% 400 == 0))$$

\]

- Examples:

- Leap years: 2004, 2008, 2012, 2016 (divisible by 4, not centuries)
- Not leap years: 1900, 2100, 2200 (century years not divisible by 400)
- Leap years: 2000, 2400 (century years divisible by 400)

- Logical Conditions in Python for Leap Year Check

- First check if the year is divisible by 100:
 - If yes, then check if divisible by 400 to confirm leap year.
- Else if year divisible by 4, then leap year.
- Else, not leap year.

- Programming Approach

- Understand the problem deeply before coding.
- Use nested conditional (if-elif-else) statements.
- Input a year and print whether it is a leap year or not.

Section 3: Example Code and Use Cases

```
year = int(input("Enter a Year: "))

if year % 100 == 0:
    if year % 400 == 0:
        print("Leap Year")
    else:
        print("Not a Leap Year")
elif year % 4 == 0:
```

```
    print("Leap Year")
else:
    print("Not a Leap Year")
```

Explanation:

- The program first takes an integer input from the user for the year.
- It then applies the Gregorian calendar rules using nested `if` statements:
 - Checks if the year is a century year (divisible by 100).
 - If century, it further checks divisibility by 400.
 - If not century, then checks divisibility by 4.
- The program outputs "Leap Year" or "Not a Leap Year" accordingly.

Use Case Examples:

- Input: `2004`

Output: `Leap Year`

- Input: `2100`

Output: `Not a Leap Year` (century year not divisible by 400)

- Input: `2400`

Output: `Leap Year` (century year divisible by 400)

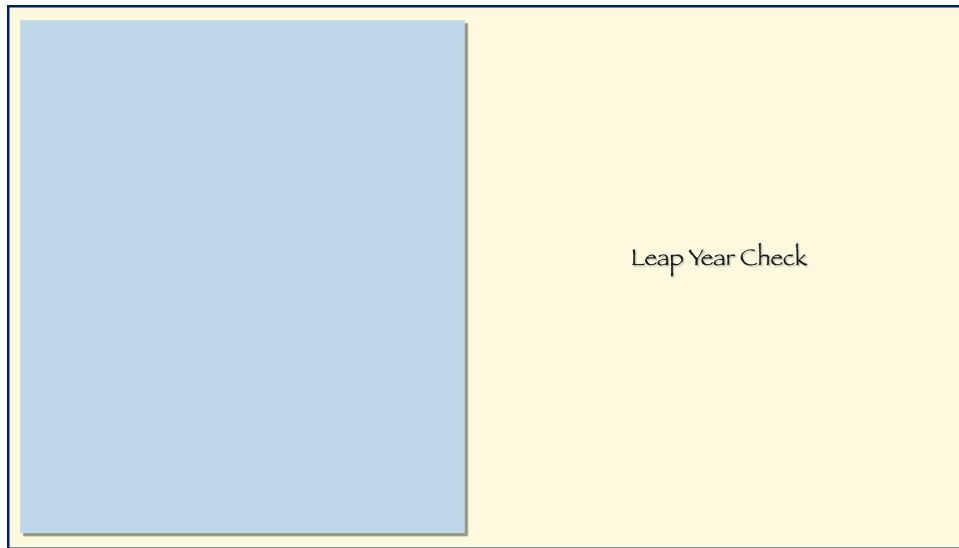
Section 4: Key Takeaways

- A leap year is a year with 366 days, where February has 29 days.

- Earth's actual revolution period (~365.242 days) necessitates occasional addition of leap days.
- The Julian calendar rule (leap year every 4 years) overcompensates, causing errors over centuries.
- The Gregorian calendar refines leap year rules with conditions based on divisibility by 100 and 400.
- Correct leap year check conditions:
 - If year divisible by 400 → leap year.
 - Else if divisible by 100 → not leap year.
 - Else if divisible by 4 → leap year.
 - Else → not leap year.
- Understanding the problem thoroughly before coding is essential for application development.
- Nested conditional statements efficiently implement the leap year rules in Python.
- Practice by testing edge cases like century years and normal leap years to validate the logic.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Leap Year Check

Challenge

Leap Year Check

Challenge

- What is a Leap Year ?

Leap Year Check
Challenge

- What is a Leap Year ?
- Julian Calendar

Leap Year Check
Challenge

- What is a Leap Year ?
- Julian Calendar
- Gregorian Calendar

Leap Year Check Challenge

- What is a Leap Year ?
- Julian Calendar
- Gregorian Calendar
- Check for Leap Year

Leap Year Check Challenge

February						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

Leap Year Check

Challenge

February

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

February

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29			

Leap Year Check

Challenge

February

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

February

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29			

Leap Year Check

Challenge

February

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

February

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29			

Leap Year

Leap Year Check

Challenge

February

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

February

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29			

365 days

Leap Year

Leap Year Check Challenge

February						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

365 days

February						
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29			

366 days

Leap Year Check

Challenge

February

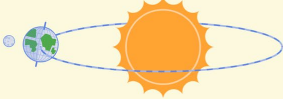
Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

February

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29			

365 days

366 days



Leap Year Check

Challenge

February

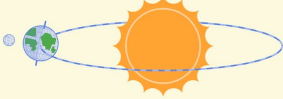
Mon	Tue	Wed	Thur	Fri	Sat	Sun
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February

Mon	Tue	Wed	Thur	Fri	Sat	Sun
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29			

365 days

366 days



365.242 days

Leap Year Check



365.242 days

Leap Year Check



365.242 days

365.25

Leap Year Check



365.242 days

365.25

365+1/4

Leap Year Check



365.242 days

365.25

365+1/4

Leap Year Check



365.242 days

365.25

365+1/4

365

Leap Year Check



365.242 days

365.25

365+1/4

365 + 1/4

Leap Year Check



365.242 days

365.25

365+1/4

365 + 1/4 * 4

Leap Year Check



365.242 days

366 = 365 + 1/4 * 4

Leap Year Check



365.242 days

$$366 = 365 + 0.25 * 4$$

Leap Year Check

 365.242 days

366 = 365 + 0.25 * 4

if year % 4 == 0:

Leap Year Check

 365.242 days

366 = 365 + 0.25 * 4

if year % 4 == 0:
'Leap Year'

Leap Year Check

 365.242 days

366 = 365 + 0.25 * 4

Julian Calendar

if year % 4 == 0:
'Leap Year'

Leap Year Check

 365.242 days

366 = 365 + 0.25 * 4

Julian Calendar

if year % 4 == 0:
'Leap~~X~~Year'

Leap Year Check

A small blue and green globe representing Earth is shown orbiting a large orange sun with rays. A thin blue elliptical line represents the orbital path.

365.242 days

Julian Calendar

if year % 4 == 0:

'Leap~~X~~Year'

Leap Year Check

 365.242 days
365.25

Julian Calendar

if year % 4 == 0:
~~'Leap Year'~~

Leap Year Check



365.242 days

365.25

$$125 * 0.008$$

Julian Calendar

if year % 4 == 0:

'Les~~X~~Year'

Leap Year Check



365.242 days

365.25

125 * 0.008 = 1

extra day

Julian Calendar

if year % 4 == 0:

'Leap~~X~~Year'

Leap Year Check



Leap Year Check

365.242 days

Leap Year Check



365.242 days

365.25

Leap Year Check



365.242 days

365.25 = 365

Leap Year Check



365.242 days

$365.25 = 365 + 0.25$

Leap Year Check



365.242 days

$365.25 = 365 + 0.25 - 0.01$

Leap Year Check



365.242 days

365.24 = 365 + 0.25 - 0.01 + 0.002

Leap Year Check



365.242 days

365.242 = 365 + 0.25 - 0.01 + 0.002

Leap Year Check



365.242 days

365.242 = 365 + 0.25 - 0.01 + 0.002

365.242 =

Leap Year Check



365.242 days

365.242 = 365 + 0.25 - 0.01 + 0.002

365.242 = 365



Leap Year Check

365.242 days

$$365.242 = 365 + 0.25 - 0.01 + 0.002$$

$$365.242 = 365 + \frac{1}{4}$$



Leap Year Check

365.242 days

$$365.242 = 365 + 0.25 - 0.01 + 0.002$$

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100}$$

Leap Year Check



365.242 days

$$365.242 = 365 + 0.25 - 0.01 + 0.002$$

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

Leap Year Check



365.242 days

$$365.242 = 365 + 0.25 - 0.01 + 0.002$$

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

Gregorian Calendar

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

if year % 4 == 0:
'Leap Year'

2004, 2008, 2012, 2016

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

if year % 100 == 0 :
'Not a Leap Year'

1900, 2100, 2200, 2300, 2500

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

if year % 100 == 0 :
if year % 400 == 0 :
'Leap Year'

2000, 2400, 2800

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

if year % 4 == 0 :

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

if :

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0:
```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0:  
    print( 'Not a Leap Year' )
```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0:
```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0:  
    if year % 400 == 0:
```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0:  
    if year % 400 == 0:  
        print( 'Leap Year' )
```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0 :  
    if year % 400 == 0 :  
        print( 'Leap Year' )  
    else:
```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0 :  
    if year % 400 == 0 :  
        print( 'Leap Year' )  
    else:  
        print( 'Not a Leap Year' )
```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0 :  
    if year % 400 == 0 :  
        print( 'Leap Year' )  
    else:  
        print( 'Not a Leap Year' )  
elif year % 4 == 0 :
```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0 :  
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    else:  
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```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0 :  
    if year % 400 == 0 :  
        print( 'Leap Year' )  
    else:  
        print( 'Not a Leap Year' )  
elif year % 4 == 0 :  
    print( 'Leap Year' )  
else:
```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
if year % 100 == 0 :  
    if year % 400 == 0 :  
        print( 'Leap Year' )  
    else:  
        print( 'Not a Leap Year' )  
elif year % 4 == 0 :  
    print( 'Leap Year' )  
else:  
    print( 'Not a Leap Year' )
```

Leap Year Check

$$365.242 = 365 + \frac{1}{4} - \frac{1}{100} + \frac{1}{400}$$

```
year = int( input( 'Enter a Year' ) )
```

```
if year % 100 == 0 :
```

```
    if year % 400 == 0 :  
        print( 'Leap Year' )
```

```
    else :  
        print( 'Not a Leap Year' )
```

```
elif year % 4 == 0 :
```

```
    print( 'Leap Year' )
```

```
else :  
    print( 'Not a Leap Year' )
```